

USEFUL FORMULAE

The Theorem of Pythagoras: $a^2 + b^2 = c^2$ if c is the length of the hypotenuse and a and b are the shorter side lengths of a right-angled triangle.

Magnitude or absolute value: $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0, \end{cases}$ which is the distance from x to 0 on the real line.

Distance between points on the real line: $|x - y|$ is the distance between x and y on the real line.

Distance between points on the plane: $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ is the distance between points $P(x_1, y_1)$ and $Q(x_2, y_2)$ on the xy -plane.

Degrees and radians: The angle 1° equals $\frac{\pi}{180}$ radians. The angle 1 radian equals $(\frac{180}{\pi})^\circ$.

Common trig values: $\sin 30^\circ = \cos 60^\circ = \frac{1}{2}$, $\sin 60^\circ = \cos 30^\circ = \frac{\sqrt{3}}{2}$, $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$,

$\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$, $\tan 45^\circ = 1$, $\cos 0 = 1$, $\cos \pi = -1$, $\sin \frac{\pi}{2} = 1$, $\sin \frac{-\pi}{2} = -1$.

Equation of a line: The line with slope m and y -intercept k has equation $y = mx + k$.

Equation of a circle: The circle with centre (a, b) and radius r has equation

$$(x - a)^2 + (y - b)^2 = r^2.$$

Area and perimeter of a circle: $A = \pi r^2$ and $P = 2\pi r$.

Volume and surface area of a sphere: $V = \frac{4}{3}\pi r^3$ and $S = 4\pi r^2$.

Quadratic formula: The quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$ has solutions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Inverse function: If $y = f(x)$ satisfies the horizontal line test then f is invertible with inverse f^{-1} , which undoes f in the sense that $x = f^{-1}(y)$.

Fractional powers and n th roots: $x^{1/2} = \sqrt{x}$, $x^{1/n} = \sqrt[n]{x}$, $x^{m/n} = (x^{1/n})^m$.

Exponential laws: $a^{-x} = \frac{1}{a^x}$, $a^x a^y = a^{x+y}$, $\frac{a^x}{a^y} = a^{x-y}$, $((a^x)^z) = a^{xz}$, $(ab)^x = a^x b^x$.

Logs and exponentials undo each other: For all x , all positive A and all positive $a \neq 1$,

$$\log_a(a^x) = x, \quad \ln e^x = x, \quad a^{\log_a A} = A, \quad e^{\ln A} = A.$$

Logarithmic laws: For all positive x and y , for all positive $a \neq 1$, and for all real k ,

$$(a) \quad \log_a(xy) = \log_a x + \log_a y, \quad \log_a(x/y) = \log_a x - \log_a y, \quad \log_a(x^k) = k \log_a x.$$

$$(b) \quad \ln(xy) = \ln x + \ln y, \quad \ln(x/y) = \ln x - \ln y, \quad \ln(x^k) = k \ln x.$$

$$(c) \quad a^x = e^{x \ln a}, \quad \log_a x = \frac{\ln x}{\ln a}.$$

Exponential growth and decay: $y = f(x) = Ae^{kx}$ where $A > 0$, and $k > 0$ for exponential growth, and $k < 0$ for exponential decay.

Logistic function: $y = f(x) = \frac{M}{1 + Ke^{-kx}}$ for constants $M, K, k > 0$.

Average rate of change: For $y = f(x)$ defined on the interval $[a, b]$, where $a < b$, the average rate of change is $\frac{f(b) - f(a)}{b - a}$.

$$\text{Derivative: } y' = \frac{dy}{dx} = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}.$$

Rules of differentiation:

$$(uv)' = uv' + vu' \quad (\text{Product Rule}) \qquad \left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2} \quad (\text{Quotient Rule})$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} \quad (\text{Chain Rule})$$

$$\text{Trick of Gauss: } \sum_{i=1}^n i = 1 + 2 + \dots + n = \frac{n(n+1)}{2}.$$

Even and odd functions: f is *even* if $f(x) = f(-x)$ and *odd* if $f(-x) = -f(x)$. If f is odd then $\int_{-a}^a f(x) dx = 0$. If f is even then $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$.

Standard integrals:

$$\begin{aligned} \int x^n dx &= \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), & \int \frac{dx}{x} &= \ln|x| + C, & \int e^x dx &= e^x + C, \\ \int \sin x dx &= -\cos x + C, & \int \cos x dx &= \sin x + C, & \int \sec^2 x dx &= \tan x + C. \end{aligned}$$

Fundamental Theorem of Calculus: If $y = f(x)$ is a continuous function defined on a closed interval $[a, b]$ and $F(x)$ is any antiderivative of $f(x)$ with respect to x then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Integration by substitution:

$$\begin{aligned} \int \left(f(u) \frac{du}{dx}\right) dx &= \int f(u) du, & \int f(g(x))g'(x) dx &= \int f(u) du, \\ \int_a^b f(g(x))g'(x) dx &= \int_{g(a)}^{g(b)} f(u) du. \end{aligned}$$